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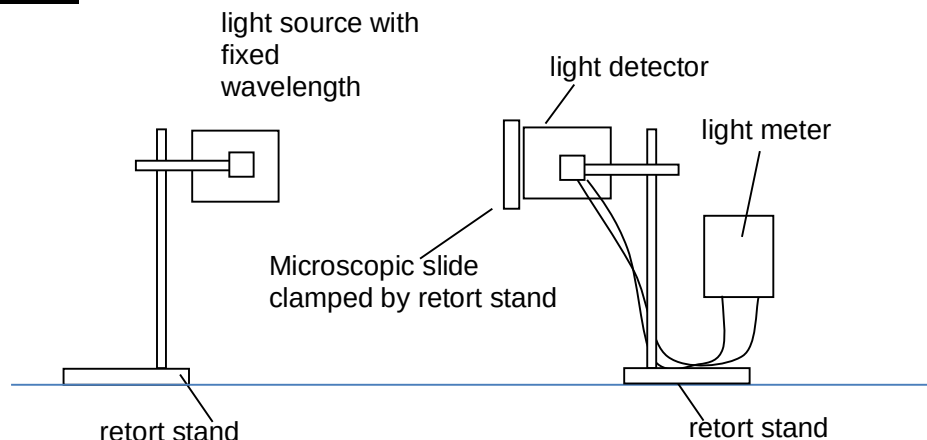
Aim: To determine the relationship between I and t .

Independent variable : t , thickness of microscopic slides

Dependent variable : I , intensity of light passing out of the microscopic slides

Controlled variables : - frequency of light
 - distance between light source and slides
 - angle of incidence of light on the slides
 - optical density of the slides

Diagram :



Procedure :

- Set up the apparatus as shown in the diagram above. Adjust the position of the light detector until the light meter shows a maximum intensity. Do not change the position of the light source and detector throughout the whole experiment.
- Measure the thickness of a piece of microscopic slide using micrometer screw gauge. Record the measurement as t .

- c) Clamp the slide in front of the light detector.
- d) Measure and record I , the reading of the light meter.
- e) Repeat the steps by adding more microscopic slides in front of the light detector to obtain six sets of readings of t and I .
- f) Assume t and I are related by the equation $I = kt^n$ where k and n are constants. Plot a graph of $\lg I$ against $\lg t$, where $n = \text{gradient}$ and $k = e^{\text{vertical-intercept}}$.

Further details :

Precautions to improve accuracy :

- Conduct the experiment in a dark room to reduce stray light measured by the light meter.
- An optical bench could be used to hold the light source, detector and slides so as to get better alignment.
- The thickness of the microscopic slide could be measured at different locations to obtain an average value of t .
- The experiment could be repeated by reducing the number of slides so that for each value of t , the intensity I could be measured twice and an average be obtained.

Safety precautions :

- Do not look directly at the intense light beam to prevent hurting eyes
- Clamp the base of the retort stands to the table with a G-clamp to avoid toppling of apparatus.